

A. All Quiet on the Western Front



Time Constraints

• **java** : 3.0 Seconds



Memory Constraints

• **java** : 256MB

A city named *Flower* on the western front, has been very quiet since some time after the deadly aftermath of war. But flowers are booming and things have started to move.

The new mayor, *Tris*, has decided to repair the country's damaged roads, which are currently in very bad condition. Tris has old maps, which have information about the previous road network. Each road of the Flower is bidirectional by default, and in the map, Tris saw an additional piece of information for each road regarding the number of vehicles that can move simultaneously through that road.

The budget is quite limited and Tris wants to maximize the total number of vehicles passing simultaneously all throughout the country. If there are N cities she wants to build exactly $(N-1)$ roads so that all the country stays connected maximizing the total number of vehicles passing simultaneously. As there are a lot of cities and roads and Tris has been very busy managing the country, she has been looking for a programmer to help her solve this problem.

Can you come forward to help her by designing a solution?

Flower can be modeled through a graph network. Flower has N cities (nodes) and E bidirectional roads(edges). In the first two lines, you will be given two integer variables N ($1 \leq N \leq 5000$) and E ($1 \leq E \leq 100000$). In the following E lines, you will be given the information regarding the roads. Each road's information holds three values A , B and C denoting an undirected edge between cities(nodes) A and B with the information C representing the total number of cars that can simultaneously pass through the road(edge) $A - > B$.

Sample Input 1

```
4 6
1 2 3
1 3 4
1 4 3
2 3 3
2 4 1
3 4 2
```

Sample Output 1

```
10
```

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